PROJECT SYNOPSIS ON

POTATO LEAF DETECTION AND DISEASE PREDICTION

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Problem Statement

Potato crops are highly vulnerable to various leaf diseases such as Early Blight and Late Blight. Early detection of these diseases is crucial to prevent crop damage and financial loss for farmers. With the increasing scale of agriculture and demand for food, manual disease detection is becoming inefficient. Leaf images in real farming environments vary in lighting, colour, background, and shape, making accurate detection challenging. Traditional rule-based or manual systems struggle to handle these variations and often produce inconsistent results.

These limitations highlight the need for an automated and reliable system that can accurately detect potato leaf diseases from images. By using image processing and deep learning techniques, it is possible to build a system that can quickly analyse leaf images and assist farmers in identifying diseases at an early stage.

Objectives

Primary Objective

To develop a fast, accurate, and easy-to-use system that can detect potato leaf diseases from leaf images using deep learning, helping farmers identify diseases early and take timely action.

Technical Objectives

1. Develop a Deep Learning Model
2. Improve Prediction Reliability
3. Enhance Efficiency

Related study

Many researchers have used image processing and deep learning techniques to detect plant diseases, especially in potato crops. Studies show that Convolutional Neural Networks (CNNs) are very effective in analysing leaf images and identifying disease patterns automatically.

For example, a study by Kumar et al. developed a CNN-based system to classify potato leaves into Early Blight, Late Blight, and Healthy categories using more than 4000 images. The model achieved around 98.5% accuracy, showing that deep learning can be very useful for early disease detection.

Other research has also explored deep learning models such as VGG, ResNet, AlexNet, and EfficientNet for plant disease classification. These models have shown high accuracy in distinguishing between healthy and diseased leaves.

Overall, these studies highlight that AI-based systems can help detect plant diseases quickly and accurately, enabling farmers to take timely action and reduce crop losses.

Methodology/approach adopted

The proposed system uses deep learning techniques to automatically detect potato leaves and identify possible diseases from images. The approach combines YOLO (You Only Look Once) for leaf detection and a Convolutional Neural Network (CNN) for disease classification.

First, the YOLO model is used to detect and locate potato leaves in an input image. This helps isolate the leaf region from the background, reducing noise and improving the accuracy of the prediction process. After the leaf is detected and extracted, the image is passed to a CNN-based classification model.

The CNN model analyses the detected leaf image and classifies it into different categories such as Early Blight, Late Blight, or Healthy. The model learns important visual features like texture, colour patterns, and spots on the leaf surface to accurately identify diseases.

Work completed so far

So far, the main parts of the Potato Leaf Detection and Disease Prediction System have been completed. A hybrid model using YOLO and TensorFlow has been developed to detect potato leaves, and a CNN model has been trained to identify whether the leaf is Healthy, Early Blight, or Late Blight. The system also includes image preprocessing to improve prediction accuracy. A React Native mobile application connected to a FastAPI backend has been built to provide real-time disease detection. Additionally, useful features such as weather updates, treatment recommendations, and regional language translation have been added to make the application more practical and user-friendly for farmers.

Future work plan

In the future, the system will be improved to make it more accurate and helpful for farmers. We plan to train the model using a larger and more diverse dataset collected from different regions, weather conditions, and crop growth stages. This will help the system recognize a wider range of potato diseases and provide more reliable predictions in real-world situations.

Another planned improvement is to add an AI assistant that can explain disease symptoms, suggest suitable treatments, answer farming-related questions, and provide guidance on disease prevention. We also aim to make the application more user-friendly by introducing features such as regional language support, offline prediction, disease history tracking, and personalized recommendations. These enhancements will make the application easier to use and more useful for farmers in their day-to-day farming activities.